Lemma 4.14 (Projection Lemma)**.** *Let $\mathcal{P} = \mathcal{P}^*$ be a bounded PI operator partitioned into three rows/columns and consider the operator inequality*

$$\begin{pmatrix} \mathcal{P}_{11} & \mathcal{P}_{12} + \mathcal{X}^* & \mathcal{P}_{13} \\ \mathcal{P}_{21} + \mathcal{X} & \mathcal{P}_{22} & \mathcal{P}_{23} \\ \mathcal{P}_{31} & \mathcal{P}_{32} & \mathcal{P}_{33} \end{pmatrix} \prec 0 \tag{4.4.5}$$

in the unstructured PI operator $\mathcal{X}$. Here and below, a self-adjoint operator $\mathcal{R}$ is coercive *if $\mathcal{R} \succeq \varepsilon I$ for some $\varepsilon > 0$, and $\mathcal{L} \prec 0$ means that $-\mathcal{L}$ is coercive. We use the PI algebra, which is closed under inversion of coercive operators. There exists a solution $\mathcal{X}$ of* (4.4.5) *if and only if*

$$\begin{pmatrix} \mathcal{P}_{11} & \mathcal{P}_{13} \\ \mathcal{P}_{31} & \mathcal{P}_{33} \end{pmatrix} \prec 0 \quad \textit{and} \quad \begin{pmatrix} \mathcal{P}_{22} & \mathcal{P}_{23} \\ \mathcal{P}_{32} & \mathcal{P}_{33} \end{pmatrix} \prec 0. \tag{4.4.6}$$

If (4.4.6) *hold, one particular solution is given by*

$$\mathcal{X} = \mathcal{P}_{32}^* \mathcal{P}_{33}^{-1} \mathcal{P}_{31} - \mathcal{P}_{21}. \tag{4.4.7}$$

Proof. (Only if) If (4.4.5) has a solution then (4.4.6) follow by canceling the first or second block row/column.

(If) Suppose that (4.4.6) holds. Then $-\mathcal{P}_{33}$ is coercive, so $\mathcal{P}_{33}^{-1}$ is a bounded PI operator. The Schur complement makes (4.4.5) equivalent to

$$\begin{pmatrix} \mathcal{P}_{11} & \mathcal{P}_{12} + \mathcal{X}^* \\ \mathcal{P}_{21} + \mathcal{X} & \mathcal{P}_{22} \end{pmatrix} - \begin{pmatrix} \mathcal{P}_{13} \\ \mathcal{P}_{23} \end{pmatrix} \mathcal{P}_{33}^{-1} \begin{pmatrix} \mathcal{P}_{31} & \mathcal{P}_{32} \end{pmatrix} \prec 0.$$

Due to (4.4.6), the negatives of the diagonal blocks are coercive. The operator $\mathcal{X}$ defined in (4.4.7) renders the off-diagonal block zero, so it is a PI solution. □

Lemma 4.15 (Projection Lemma)**.** *For bounded PI operators $\mathcal{A} : \mathcal{H} \to \mathcal{Y}_A$, $\mathcal{B} : \mathcal{H} \to \mathcal{Y}_B$ and $\mathcal{P} = \mathcal{P}^* : \mathcal{H} \to \mathcal{H}$, consider*

$$\mathcal{A}^* \mathcal{X} \mathcal{B} + \mathcal{B}^* \mathcal{X}^* \mathcal{A} + \mathcal{P} \prec 0 \tag{4.4.8}$$

in the unstructured PI operator $\mathcal{X} : \mathcal{Y}_B \to \mathcal{Y}_A$. For the converse below, assume that the block spaces used in the proof admit PI coordinates, that the coordinate map $\mathcal{S}$ is PI, and that $\mathcal{S}^*\mathcal{S}$, $\mathcal{A}_c^*\mathcal{A}_c$ and $\mathcal{B}_c^*\mathcal{B}_c$ are coercive. The admissible PI class is closed under sums, products and adjoints; zero-dimensional blocks are omitted.

Then (4.4.8) *has a solution if and only if*

$$\exists\, \delta > 0 : \quad \mathcal{A}x = 0 \text{ or } \mathcal{B}x = 0 \implies \langle x, \mathcal{P}x \rangle \le -\delta \|x\|^2. \tag{4.4.9}$$

If $\mathcal{A}_\perp$ and $\mathcal{B}_\perp$ are isometric parameterizations of $\ker \mathcal{A}$ *and* $\ker \mathcal{B}$*, respectively,* (4.4.9) *is equivalent to*

$$\mathcal{A}_\perp^* \mathcal{P} \mathcal{A}_\perp \prec 0 \quad \textit{and} \quad \mathcal{B}_\perp^* \mathcal{P} \mathcal{B}_\perp \prec 0. \tag{4.4.10}$$

Proof. The proof of *only if* is immediate. If $\mathcal{A}x = 0$ or $\mathcal{B}x = 0$, the cross terms vanish, so (4.4.8) implies (4.4.9). This direction needs no additional coordinate or coercivity assumptions.

For proving *if*, define

$$\mathcal{H}_3 = \ker \mathcal{A} \cap \ker \mathcal{B}, \qquad \mathcal{H}_1 = \ker \mathcal{A} \ominus \mathcal{H}_3,$$
$$\mathcal{H}_2 = \ker \mathcal{B} \ominus \mathcal{H}_3, \qquad \mathcal{H}_4 = (\ker \mathcal{A} + \ker \mathcal{B})^\perp.$$

Let $\mathcal{S} : \bigoplus_{j=1}^4 \mathcal{H}_j \to \mathcal{H}$ be $\mathcal{S}(x_1, x_2, x_3, x_4) = x_1 + x_2 + x_3 + x_4$. Coercivity of $\mathcal{S}^*\mathcal{S}$ makes $\mathcal{S}$ bounded below with closed range. Its range is dense by construction, so $\mathcal{S}$ is boundedly invertible. Instead of (4.4.8), consider the equivalent inequality

$$(\mathcal{A}\mathcal{S})^* \mathcal{X} (\mathcal{B}\mathcal{S}) + (\mathcal{B}\mathcal{S})^* \mathcal{X}^* (\mathcal{A}\mathcal{S}) + \mathcal{S}^* \mathcal{P} \mathcal{S} \prec 0. \tag{4.4.11}$$

Now $\mathcal{A}\mathcal{S}$ and $\mathcal{B}\mathcal{S}$ have the structure $(0\ \mathcal{A}_2\ 0\ \mathcal{A}_4)$ and $(\mathcal{B}_1\ 0\ 0\ \mathcal{B}_4)$. The operators $\mathcal{A}_c = (\mathcal{A}_2\ \mathcal{A}_4)$ and $\mathcal{B}_c = (\mathcal{B}_1\ \mathcal{B}_4)$ are injective. Coercivity of $\mathcal{A}_c^*\mathcal{A}_c$ and $\mathcal{B}_c^*\mathcal{B}_c$ gives the bounded PI left inverses

$$\mathcal{A}_c^\dagger = (\mathcal{A}_c^*\mathcal{A}_c)^{-1}\mathcal{A}_c^*, \qquad \mathcal{B}_c^\dagger = (\mathcal{B}_c^*\mathcal{B}_c)^{-1}\mathcal{B}_c^*.$$

Thus the equation

$$(\mathcal{AS})^*\mathcal{X}(\mathcal{BS}) = \begin{pmatrix} 0 \\ \mathcal{A}_2^* \\ 0 \\ \mathcal{A}_4^* \end{pmatrix} \mathcal{X} \begin{pmatrix} \mathcal{B}_1 & 0 & 0 & \mathcal{B}_4 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ \mathcal{Z}_{21} & 0 & 0 & \mathcal{Z}_{24} \\ 0 & 0 & 0 & 0 \\ \mathcal{Z}_{41} & 0 & 0 & \mathcal{Z}_{44} \end{pmatrix}$$

has a PI solution for arbitrary bounded PI blocks $\mathcal{Z}_{21}, \mathcal{Z}_{24}, \mathcal{Z}_{41}, \mathcal{Z}_{44}$: take

$$\mathcal{X} = (\mathcal{A}_c^\dagger)^* \begin{pmatrix} \mathcal{Z}_{21} & \mathcal{Z}_{24} \\ \mathcal{Z}_{41} & \mathcal{Z}_{44} \end{pmatrix} \mathcal{B}_c^\dagger.$$

With $\mathcal{Q} := \mathcal{S}^*\mathcal{P}\mathcal{S}$ partitioned accordingly, (4.4.11) hence reads as

$$\begin{pmatrix} \mathcal{Q}_{11} & \mathcal{Q}_{12} + \mathcal{Z}_{21}^* & \mathcal{Q}_{13} & \mathcal{Q}_{14} + \mathcal{Z}_{41}^* \\ \mathcal{Q}_{21} + \mathcal{Z}_{21} & \mathcal{Q}_{22} & \mathcal{Q}_{23} & \mathcal{Q}_{24} + \mathcal{Z}_{24} \\ \mathcal{Q}_{31} & \mathcal{Q}_{32} & \mathcal{Q}_{33} & \mathcal{Q}_{34} \\ \mathcal{Q}_{41} + \mathcal{Z}_{41} & \mathcal{Q}_{42} + \mathcal{Z}_{24}^* & \mathcal{Q}_{43} & \mathcal{Q}_{44} + \mathcal{Z}_{44} + \mathcal{Z}_{44}^* \end{pmatrix} \prec 0. \tag{4.4.12}$$

Since

$$\ker(\mathcal{AS}) = \operatorname{im} \begin{pmatrix} I & 0 \\ 0 & 0 \\ 0 & I \\ 0 & 0 \end{pmatrix}, \qquad \ker(\mathcal{BS}) = \operatorname{im} \begin{pmatrix} 0 & 0 \\ I & 0 \\ 0 & I \\ 0 & 0 \end{pmatrix},$$

the hypothesis (4.4.9) amounts to

$$\begin{pmatrix} \mathcal{Q}_{11} & \mathcal{Q}_{13} \\ \mathcal{Q}_{31} & \mathcal{Q}_{33} \end{pmatrix} \prec 0, \qquad \begin{pmatrix} \mathcal{Q}_{22} & \mathcal{Q}_{23} \\ \mathcal{Q}_{32} & \mathcal{Q}_{33} \end{pmatrix} \prec 0.$$

By Lemma 4.14, since $-\mathcal{Q}_{33}$ is coercive and therefore $\mathcal{Q}_{33}^{-1}$ is PI, choose

$$\mathcal{Z}_{21} = \mathcal{Q}_{23}\mathcal{Q}_{33}^{-1}\mathcal{Q}_{31} - \mathcal{Q}_{21}$$

to make the negative of the leading 3×3 block in (4.4.12) coercive. If $\mathcal{H}_3 = \{0\}$, take $\mathcal{Z}_{21} = -\mathcal{Q}_{21}$ instead. The blocks $\mathcal{Z}_{41}$ and $\mathcal{Z}_{24}$ can be arbitrary bounded PI operators. After fixing them, write (4.4.12) as

$$\begin{pmatrix} \mathcal{R} & \mathcal{F} \\ \mathcal{F}^* & \mathcal{Q}_{44} + \mathcal{Z}_{44} + \mathcal{Z}_{44}^* \end{pmatrix} \prec 0, \qquad \mathcal{R} \prec 0.$$

Choose $\mathcal{Z}_{44} = -\alpha I$ with

$$2\alpha > \|\mathcal{Q}_{44}\| + \|\mathcal{F}\|^2\|\mathcal{R}^{-1}\|.$$

The negative of the Schur complement is coercive. This is the operator counterpart of the original proof's application of Lemma 4.13. Coercivity of $-\mathcal{R}$ ensures that $\mathcal{R}^{-1}$ is bounded and PI. Pulling back through the boundedly invertible $\mathcal{S}$ proves (4.4.8). □